

Commands and Codes

Demo: Demo: Table Design - Distribution Style | Queries

```
create schema table_design_demo;

SET search_path = table_design_demo,public;

-- checking the sortkey for users
select "column", type, distkey
from pg_table_def where tablename = 'users' and schemaname='public';

select * from svv_table_info s where s. schema='public' and s.table ='users';

-- USERID is defined as the DISTKEY column

-- checking the disk usage, distribution
select slice, col, num_values as rows, minvalue, maxvalue
from svv_diskusage
where name='users' and tbl = (select table_id from svv_table_info s where s.schema='public' and
s.table ='users')
and col=0
and num_values>0
order by slice, col;

-- table is evenly distributed, same range,
-- col is userid

-- state as the dist key
create table userskey distkey(state) as select * from users;

-- checking the disk usage with state as distkey
select slice, col, num_values as rows, minvalue, maxvalue
from svv_diskusage
where name='userskey' and tbl = (select table_id from svv_table_info s where
s.schema='table_design_demo' and s.table ='userskey')
and col=0
and num_values>0;

-- not evenly distributed
-- even e. g.
create table userseven diststyle even as
select * from users;

select slice, col, num_values as rows, minvalue, maxvalue
from svv_diskusage
where name='userseven' and tbl = (select table_id from svv_table_info s where
s.schema='table_design_demo' and s.table ='userseven')
and col=0
and num_values>0;

-- all example
create table usersall diststyle all as
select * from users;
```

```

select slice, col, num_values as rows, minvalue, maxvalue
from svv_diskusage
where name='usersall' and tbl = (select table_id from svv_table_info s where
s.schema='table_design_demo' and s.table = 'usersall')
and col=0
and num_values>0;

-- auto example
create table usersauto as
select * from users;

select slice, col, num_values as rows, minvalue, maxvalue
from svv_diskusage
where name='usersauto' and tbl = (select table_id from svv_table_info s where
s.schema='table_design_demo' and s.table = 'usersauto')
and col=0
and num_values>0;

select * from svv_table_info s where s. schema='table_design_demo' and s. table = 'usersauto';

```

Demo: Table Design - Sort Key | Queries

```

---- sort
select "column", type, sortkey
from pg_table_def where tablename = 'usersauto';

-- update the sort key
create table users_custom_sort sortkey (firstname, lastname) as
select * from users;

select "column", type, sortkey
from pg_table_def where tablename = 'users_custom_sort';

explain SELECT firstname, lastname, total_quantity
FROM (SELECT buyerid, sum(qtysold) total_quantity
      FROM sales
      GROUP BY buyerid
      ORDER BY total_quantity desc limit 10) Q, table_design_demo.users_custom_sort U
WHERE Q.buyerid = U.userid
ORDER BY Q.total_quantity desc;

-- explain functions
-- cost - 1st value to return the first row, 2nd value to complete the whole operation

```

Demo: Table Design - Compression | Queries

```

-- compression
select "column", type, encoding
from pg_table_def where tablename = 'usersauto' and schemaname='table_design_demo';

```

```
ANALYSE COMPRESSION usersauto;
```

```
create table users_custom_compression(  
  userid integer not null distkey sortkey,  
  username char(8) encode zstd,  
  firstname varchar(30) encode zstd,  
  lastname varchar(30) encode zstd,  
  city varchar(30),  
  state char(2),  
  email varchar(100),  
  phone char(14),  
  likesports boolean,  
  liketheatre boolean,  
  likeconcerts boolean,  
  likejazz boolean,  
  likeclassical boolean,  
  likeopera boolean,  
  likerock boolean,  
  likevegas boolean,  
  likebroadway boolean,  
  likemusicals boolean);  
  
-- inserting records from users  
insert into users_custom_compression  
select * from users  
  
select "column", type, encoding  
from pg_table_def where tablename = 'users_custom_compression' and schemaname='table_design_demo';  
  
ANALYSE COMPRESSION users_custom_compression;
```